



Figure 10: Diagram of the decoding protocol. When Alice relinquishes control of the impurity Bob performs a reversal of the dynamics up to that point except he only has access to the medium, M , and the impurity discarded by Alice in the refresh step, D .

Applying this relation to the relevant channels of Bob and Charlie then yields

$$I(A \rightarrow \text{Charlie}) = -I(A \rightarrow \text{Bob}). \quad (14)$$

This equivalence implies that $I(A \rightarrow \text{Charlie})$ must also witness the backflow transition when v_d is varied. In contrast to Bob, Charlie has no information about Alice's qubits in the Markovian regime (when Bob can decode perfectly). While, in the regime of operator Non-Markovianity, Charlie has partial information about Alice's qubits.

5.2.1 Teleportation decoder

As before, we can construct a decoding protocol that reconstructs this partial information as shown in Figure 11. In that figure, the decoder is shown in green, and we show the diagram for the corresponding fidelity, $\tilde{\mathcal{F}}$, in Appendix C. There is a clear similarity with the Hayden-Preskill protocol decoder proposed by Yoshida and Kitaev [10]; in both cases a Bell measurement is performed to facilitate the teleportation of Alice's information to Charlie's qubits. The teleportation that Charlie hopes to achieve in this case is indicated by the blue arrow in Figure 11. The success of the teleportation improves when the bell measurement probability is lower (see Appendix C). Given that R and A_R are initially in a Bell state, the decoder has the worst fidelity when $\text{Tr}_M[U_2 \rho_M(t_1)]$ is unitary, i.e. U_2 and U_2^* do nothing to disentangle R and A_R .

As in the previous protocol, we can relate Charlie's decoder fidelity, $\tilde{\mathcal{F}}$ to the coherent information (details in Appendix C):

$$I(A \rightarrow \text{Charlie}) = 1 + \log_2 \tilde{\mathcal{F}}. \quad (15)$$

The logarithm of the fidelity, therefore also observes the backflow transition. Note, that while Bob's decoder works perfectly in the Markovian phase, Charlie's decoder only partially succeeds in the non-Markovian phase. This is the main difference with the Hayden-Preskill protocol, and highlights the fact that the scrambling of operators back on to the impurity is not maximal, as is assumed for the black hole in the Hayden-Preskill protocol.